

BARRIERS TO INNOVATION DIFFUSION IN INDUSTRIAL NETWORKS: A SYSTEMATIC COMBINING APPROACH

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ABSTRACT

Over the past 50 years, a substantial interest has been put to research on how innovation spreads within social networks over time (see Rogers, 1962, 2010). Our initial aim was to examine innovation diffusion in industrial networks. We operationalized the research through a case study of an advertising network by using systematic combining as the approach (Dubois & Gadde, 2002, 2014). From the initial focus of innovation diffusion, the rematching of data and theory led us to focus on the barriers of innovation diffusion. By doing so, we found out that multilevel strategizing appears to be an important phenomenon in understanding dynamics of innovation diffusion within industrial networks. Specifically, strategizing occurs in two levels: (1) the groups within the network compete for position, and (2) actors within a group compete for position by trying to differentiate themselves from other group actors. A strategic

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mismatch between the two levels leads the network to become decelerated or even static in diffusing new innovations (Abrahamsen, Henneberg, & Naudè, 2012). Uncovering these findings would not have been possible without the use of systematic combining and the constant matching between theoretical and empirical domains.

Keywords: Systematic combining; industrial networks; innovation diffusion; advertising

INTRODUCTION

Business network studies focus on examining how various economic activities occur within a networked structure (Kogut, 2000; Powell, 1990; Tortoriello & Krackhardt, 2010). An especially rich and growing body of network research focuses on understanding network dynamics and exploring how network structures, relations between actors, and resource constellations change over time. The network structure is seen as a continually evolving interplay between various actors of the net (Gulati, Nohria, & Zaheer, 2000), where the role of a single actor also undergoes dynamic change. These changes, both in the network structure and in the role of the actors, have emerged through the changes in the substance and function of the network (Håkansson & Snehota, 1995).

Recently, network and innovation scholars have focused on understanding the network dynamics that drives the diffusion of innovation within networks (Ceci & Iubatti, 2012; Kogut, 2000). Innovation diffusion is the process through which innovation spreads within a specific social system over a period of time (Peres, Muller, & Mahajan, 2010; Rogers, 1962, 2010). Previous studies that focus on innovation diffusion within business networks have identified various structural factors that increase the likelihood of adaptation and innovation diffusion within networks, such as high levels of centralization, small numbers of actors, and the multidimensionality of networks (Ceci & Iubatti, 2012; Cooke, 2001; Valente, 1995).

However, as extant research focuses on demonstrating innovation diffusion through computerized simulations (e.g., Abrahamson & Rosenkopf, 1993; Choi, Sang-Hoon, & Lee, 2010), industrial innovation diffusion research lacks real-life examples of how innovations diffuse. Studies that examine innovation diffusion in a real-life setting have mainly focused on innovation diffusion in strategic entrepreneurial nets aimed at improving the effectiveness of innovation performance (e.g., Cooke, 2001). The

application of these findings to other types of network settings is somewhat problematic, as the network type can have a substantial effect on the manner in which innovations diffuse (Midgley, Morrison, & Roberts, 1992). Thus, more understanding of innovation diffusion in different real-life network settings is needed.

The study was originally designed to provide a framework of the innovation diffusion process in an existing business network setting. As we found that innovations were not diffusing within the network we studied, the focus shifted from framing the industrial innovation diffusion process into extending our understanding of possible barriers to innovation diffusion within the net. Specifically, we became interested in uncovering both individual actor-level and group-level factors (Wilke & Ritter, 2006) that decelerate the innovation diffusion in this context, especially strategizing (Gadde, Huemer, & Håkansson, 2003).

We operationalize the research as a case study (see, e.g., Eisenhardt, 1989; Yin, 2003) based on 6 interviews of industry experts and 16 interviews with participants of different actor groups. Our research design follows the systematic combining approach, which is based on abductive logic and continuous iteration between the theory and the empirical data (Dubois & Gadde, 2002). As systematic combining is a novel approach for innovation diffusion research, we explain in the methodology section how systematic combining is applied into our research.

The study here focuses on advertising networks in Finland between 2010 and 2012. During the three-year period, possibilities to advertise in tablet devices were emerging. This situation provides a rich setting to study innovation diffusion at the intersection of traditional print advertising and rapidly expanding digital advertising.

THEORETICAL BACKGROUND

According to the view of the Industrial Marketing and Purchasing (IMP) research group, markets are complex networks of dependent exchange relationships (Axelsson, 1992; Dyer & Singh, 1998; Halinen, Salmi, & Havila, 1999). According to Möller and Svahn (2003) networks are value systems that seek to offer end customers “whole products” through knowledge transfer and joint value creation between the actors. Specifically, three types of value systems can be identified, stable and well-defined value systems, established value systems, and emerging value systems.

In network-related studies, focus has especially been on network interactions between multiple actors (Håkansson & Snehota, 1995). Understanding the spatial and interlinked relationships between actors within a business network is seen a key requirement for understanding any type of a network (Henneberg, Naudé, & Mouzas, 2010).

The basic assumption in the industrial network approach is that analyzing these relationships requires examination of actors, resources, and activities within the focal network (Håkansson & Snehota, 1995). Additionally, Wilke and Ritter (2006) have suggested a distinction between structural and actor-level analysis of network interactions. The structural level analysis consists of examination of network relationships between multiple actors within the focal network. On the other hand, in the actor level of analysis, the organizations in the focal network are examined (Wilke & Ritter, 2006).

During recent years, research has investigated innovation diffusion within the industrial network context (Ceci & Iubatti, 2012; Kogut, 2000). According to innovation diffusion theory, innovation diffusion is the process of the penetration of new products and services into social networks, which is driven by social influences (Peres et al., 2010; Rogers, 2010). These social influences may include social interdependencies of all kinds within the network (Goldenberg, Libai, & Muller, 2010), such as various groups of actors within an industrial network.

With regards to innovation diffusion process, current network and innovation diffusion studies have already demonstrated how innovation diffuses within simulated environments (Abrahamson & Rosenkopf, 1993; Choi et al., 2010). For instance, Choi et al. (2010) find that different cliques between groups of actors within the net have crucial roles in defining how innovations diffuse. Although innovation diffusion has been researched extensively through computerized modeling, we believe that this field of study has much more to offer in terms of describing the innovation diffusion in both real-life and industrial network contexts.

Our study builds on both industrial networks theory and innovation diffusion theory. As we are interested in how innovations are diffusing in real-life settings where networks are changing, we apply the actors, resources, and activities (ARA) model (Håkansson, 1987; Håkansson & Snehota, 1995) to unravel the structures and between-actor relationships in the industrial diffusion process. In order to understand multiple perspectives of possible barriers of innovation diffusion, we explore both firm level and group-level activities within the network (Wilke & Ritter, 2006). In doing

so, we focus on strategizing behavior (Gadde et al., 2003) to cover simultaneous strategizing in these two levels.

METHOD

In order to examine innovation diffusion within existing business network and the dynamics of the participating actors, the resources ties, and the activity links related to innovation diffusion, we selected a case study method. A case study approach (e.g., Stake, 1995; Yin, 2003) is considered to be appropriate for two main reasons. First, according to Yin (2003) a case study approach is a useful method for exploring complex phenomena within its context. Second, it is the recommended method for understanding dynamics of complex network systems with unclear boundaries (Halinen & Törnroos, 2005).

Within the case study framing we utilize a systematic combining approach (Dubois & Gadde, 2002, 2014). According to Dubois and Gadde (2002, p. 556) systematic combining is defined as “a non-linear, path-dependent process of combining efforts with the ultimate objective of matching theory and reality.” When the systematic combining approach is utilized, theory and empirical data are continuously adapted to fit with each other through the processes of traveling back and forth between theoretical realm, empirical realm, and case analysis. As theory, empirical work, and analysis are evolving simultaneously, systematic combining is exceptionally useful for theory development (Dubois & Gadde, 2002).

The stages through which this research process evolves can be identified in retrospect (Dubois & Gadde, 2002). Fig. 1 shows how the study combines both empirical fieldwork and existing conceptual insights to broaden our understanding of innovation diffusion in industrial networks. The process is highly iterative as we move back and forth from theory to empirics numerous times.

Fig. 1 also indicates how our understanding of the phenomenon progresses. In Step 1, our aim was to frame innovation diffusion process. First we conducted six interviews with industry experts to gain a deeper understanding of an advertising network within the context of industrial networks. In this step, we followed the long interview approach advocated by McCracken (1988). Therefore, we first focused on grand tour questions, followed by specific questions of the categories that the interviewees had

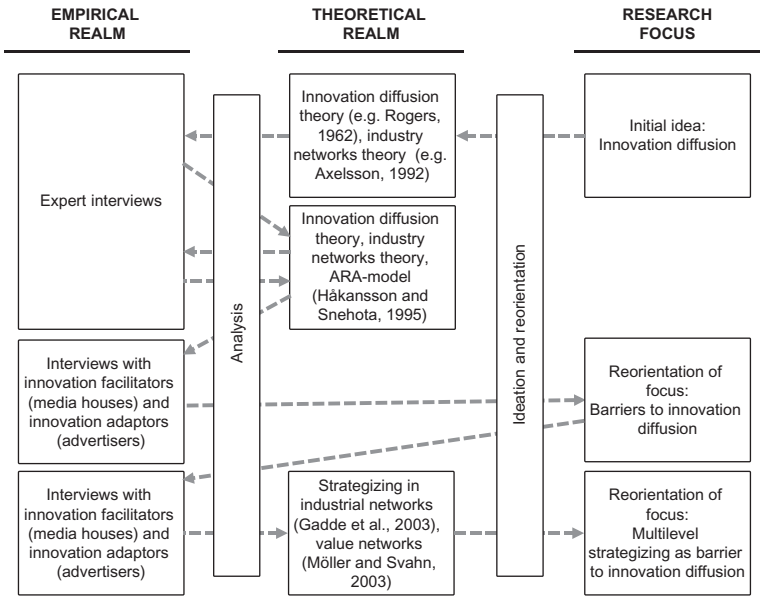


Fig. 1. The Systematic Combining Research Process.

identified. This enables us to get an overview of how Finnish advertising industry functions.

After gaining an overview of how the advertising industry functions according to experts, we returned to the theoretical realm and applied the ARA-model (Håkansson, 1987; Håkansson & Snehota, 1995) to analyze the network. This enabled us to distinguish the kinds of actors, the activities they conduct, and what kind of activity bonds there could be. We then consulted industry experts to validate our identification of different actors within the network.

The next step in our research process was to conduct seven semi-structured interviews with the different actors in the network. These actors are three innovation facilitators (media houses) and four innovation adaptors (advertisers). At this point, we faced a dilemma: Innovations were not diffusing among actor groups in the advertising network.

As neither innovation facilitators nor innovation adopters appeared complacent about the diffusion of the innovation on their side, we turned our attention toward innovation intermediaries that function between the facilitators and adapters and their role in decelerating the diffusion of

innovations. Therefore, our focus was shifted from framing innovation diffusion process to examining barriers to innovation diffusion, by exploring conflicts between and within actor groups. Thus, our research was redirected from studying the process of innovation diffusion into explaining the deeper reasons behind the lack of innovation diffusion.

In the final step, we conducted nine semi-structured interviews with various actor groups within the network, focusing on innovation intermediaries (media, advertising, and digital agencies) that were previously identified. By iterating the research data and theoretical realm, we were able to distinguish how different actors and groups of actors decelerate the adoption of innovations. This enabled us to first, produce a description of the phenomenon from which we then started building the explanation (Pentland, 1999).

All the interviews were tape recorded and transcribed later. Table 1 depicts the persons interviewed in each stage, their position, the group they belong to, and their advertising network experience.

FINDINGS

Based on our empirical research, we define various groups of actors and their activities in the network. The advertising network consists of five broad groups of actors: (1) media houses that offer media space, (2) media agencies that sell communications planning and media space, (3) advertising agencies that produce advertisements, (4) new digital agencies that provide specialist services related to new digital media content, and (5) advertisers that purchase advertisements and ad space. On the activity level, intermediary actor groups have overlapping activities as media agencies, advertising agencies, and digital agencies partake in the planning and execution of advertising campaigns. Fig. 2 illustrates how these actor groups are interlinked with each other in the advertising network structure.

Before the digitalization of media industry, advertising network consisted of only four actor groups, namely media houses, media agencies, advertising agencies, and advertisers. The value creation in this network was created through joint efforts of all actor groups, where advertising agencies were the leading actors. Following the categorization of Möller and Svahn (2003), this represents a stable and well-defined value system, where value activities, actors, technologies, and procedure are specified and well known.

Table 1. Interviewees, Their Position and Background.

Informant No.	Position	Group	Advertising Network Experience
1	CEO	Industry expert	Unknown ^a
2	CEO	Industry expert	Unknown ^a
3	Partner	Industry expert	Unknown ^a
4	Partner	Industry expert	Unknown ^a
5	CEO	Industry expert	Unknown ^a
6	CEO	Industry expert	Unknown ^a
7	CEO	Media house	7 years
8	Planning Director	Media house	2 years
9	Team Manager	Media house	7 years
10	Brand Marketing Manager	Advertiser	7 years
11	Area Manager	Advertiser	1 year
12	Online-Dialog Director	Advertiser	Less than a year
13	Marketing Manager	Advertiser	8 years
14	Head of Digital Advertising	Digital agency	Less than a year
15	Account and Sales Manager	Digital agency	Less than a year
16	Creative Director	Digital agency	Less than a year
17	Producer	Digital agency	1 year
18	CEO	Advertising agency	5 years
19	Creative Strategist	Advertising agency	2 years
20	Account Executive	Media agency	4 years
21	Digital Director	Media agency	1 year
22	CEO	Media agency	2.5 years

^aExperience of industry experts is unknown as questions related to advertising network experience were not included in early stages of our research.

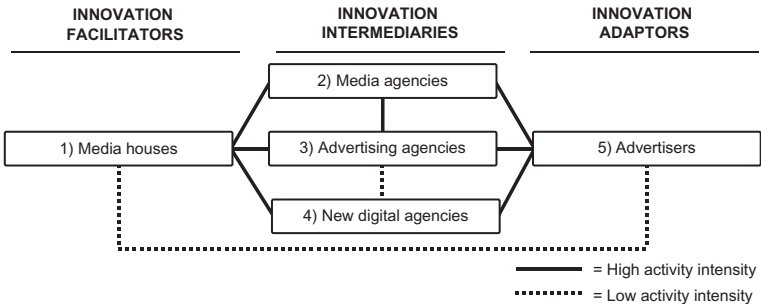


Fig. 2. Illustration of the Finnish Advertising Network.

The new digital agencies emerged during 2000s when neither the advertising agencies nor media agencies had the ability to produce digital advertisements. As such this did not pose direct challenges to the positions of the conventional actors in the existing network structure, as the digital agencies initially assumed a subcontractor position and they were unable to gain resources and conduct activities that would have threatened the positions of the advertising agencies. As one of our interviewees stated on the past roles of different actors:

The 1980s was a goldmine for the advertising agencies and still in the 1990s TV [advertising] was like a money-printing machine. (Creative Director, informant no. 16)

As the marketplace of digital marketing grew at the beginning of the new millennium, the network positions of digital agencies became reconfigured into a more central role in the Finnish advertising network. These developments have effectively separated advertising and digital agencies into distinct network actors. This dynamic is largely due to major cuts in firms' advertising budgets as a result of the economic downturn and due to the introduction of new technologies. These two actor categories thus emerged into distinct groups of network actors that compete with each other for customers and contracts. However, this competitive setup with regards to activities of different actors is not very clear for different actor groups in the network. Due to these changes in technology, value-creating activities, and actors, the network is incrementally developing, representing an established value system (Möller & Svahn, 2003). These changes are well exemplified by the following quote from one of the interviewees:

... We used to have clear roles that the advertising agency does the visual ... and media agency makes the choice on media channels ... and now when the digital agencies have entered the field, the digital has kind of stirred the soup ... (Planning Director, informant no. 8)

With regards to innovation diffusion, media houses have taken the primary role of developing and facilitating new advertising innovations within the network. This is largely due to the falling subscription rates that need to be mitigated through new forms of generating cash flow. On the other hand, all of the interviewed managers from the advertiser group say that they are willing to launch pilots with new advertisement techniques, such as tablet advertising, if these become available.

Despite the fact that there are both active innovation facilitator group and potential innovation adaptor group within the established value

system, advertising innovations are not spreading in this system. This situation appears to stem from conflicts between the three intermediary groups, namely digital, media, and advertising agencies. Actors within these groups are not able to share resource ties and activity links together mainly because of group-level cliques and individual-level positioning.

First, two groups of network actors, media and advertising agencies, try to defend their groups' existence. The majority of interviewed media and advertising agencies highlight the importance of both the quality of the creative idea and the use of traditional advertising measures, which were central value-creating activities in the previous stable value system. Creativity has been the core activity of advertising agencies, while the core activity of media agencies has been measurement. This juxtaposition is exemplified well by the following two quotes, first from a media agency and second from an advertising agency:

We have to get the measurement right [in digital context]. It has to have the right KPIs [key performance indicators] and we have to measure them. (CEO, informant no. 22)

This is very simple equation. Effectiveness of the advertisement [idea of the advertisement] times the number of contacts equal the effectiveness of the advertising campaign. If you have a bloody good idea and you reach 50.000 people with it, it is better than having a decent idea and reach 2.500.000 people. (CEO, informant no. 18)

On the other hand, digital agencies emphasize the new features of digital channels, such as personalization and the inaccuracy of traditional measurement. This differs substantially from the perspective of both advertising agencies and media agencies. The new way of how value is created in the existing value system is illustrated by a digital agency representative:

These kinds of digital mediums that identify the users are increasing, where you can automate the measurement that gives the log [of user data] out which is way more accurate ... Idea of brand and its functionality can be conveyed more clearly through interactivity and rich content [in digital media] than through a single picture or text [in traditional media]. (Creative Director, informant no. 16)

The intermediary groups therefore blatantly compete with each other by emphasizing their strengths and playing down the role of the other groups. This creates an "us against them" strategizing at the group level. In doing so, advertising agencies and media agencies draw from the old stable and well-defined value system, whereas digital agencies draw from the established value system under incremental change. This mismatch appears to explain why these groups compete against each other and, on the group level, hinder the diffusion of innovations.

Second, the firms in the intermediary groups compete within the group for positions and network influence. Within the group, individual actors' strategizing takes the form of, for instance, positioning themselves by being a frontrunner or a different kind of agency. This was highlighted by a media agency representative, who noted the following:

[Usually] media agencies thinking starts from a budget that has been decided to be allocated to a certain [bought] media. [Our] engagement thinking starts from the point that companies exist for their customers and the smartest marketing is to first identify and utilize company's own channels and then think what role does bought media have. (CEO, informant no. 22)

These networked groups pose similar intra-group activity readiness by emphasizing specific and distinct aspects of advertising that comply with their core activities. Together, the dynamics of these strategizing activities within and between these groups creates a suboptimal situation for innovation diffusion. As one media house manager said:

In Finland 82% of all media advertisement is conducted via media agencies. We have conducted 5 years of digital advertising through our 11 magazine brands. So guess how many digital advertisements we have sold via media agencies? One. (CEO, informant no. 7)

Multilevel strategizing is an important phenomenon in understanding networks where there are numerous actors in several distinct competing groups that have overlapping roles when the value system shifts from being stable into being an incrementally changing one. These groups within the network compete for position. Simultaneously, actors within a group compete for position by trying to differentiate themselves from other group actors.

DISCUSSION

This research provides an example of how multilevel strategizing conflicts lead a network to decelerated or even static position in diffusing innovations. Thus, it extends the sparse empirical knowledge of the factors that may decelerate innovation diffusion in industrial networks.

According to [Abrahamsen, Henneberg, and Naudé \(2012\)](#) firms aim to change the network in a desired direction by altering how the role of the focal firms is interpreted. Our study highlights that firms have a dual role in trying to alter how they are identified. More specifically, firms strive to belong to a group, but simultaneously try to position themselves within the groups as being different.

Interestingly, search for a dominant position in the network becomes a dual phase process. First, the groups within the network compete for position. Second, actors within a group compete for position by trying to differentiate themselves from other group actors. Due to the existing network structure and functions, actors from different competing groups have problems in generating activity links and mutual understanding that would benefit both of the actors in their own category (Munksgaard, Clarke, Storvagn, & Erichsen, 2012). This results in strategic mismatch between the groups, which leads the network to become decelerated or even static in diffusing new innovations (Abrahamsen et al., 2012).

When looking at the network from a value system perspective (Möller & Svahn, 2003), change from a stable and well-defined value system into an incrementally changing value system can create conflicts between actors. Furthermore, when new actors provide alternative ways of creating value in the new value system, the propensity for conflicts is accelerated. This may partly explain why innovations are not fully adopted even though every actor group in the network acknowledges them.

Methodologically, this research also provides suggestions. The processes of abduction are oftentimes hidden within journal articles, as has been noted by Dubois and Gadde (2014). However, in the present study the process helped in discovering a new perspective into the phenomenon and therefore followed the original suggestion of Dubois and Gadde (2002) in using the method. We strongly suggest that these kinds of processes should be made transparent as they give depth, complexity, and credence to business reality. Simultaneously this requires researchers using systematic combining to outline how the method was used and how the movement between empirical and theoretical domains was made to reach conclusions. Making this process explicit when reporting the study, in our opinion, gives the study its methodological rigor.

Systematic combining as an approach provides tools for understanding deep and complex structures and does not shy away from unanticipated empirical findings (Dubois & Gadde, 2002). These kinds of unanticipated findings are easily left outside the scope of multiple case studies that strive for analytical generalization (Eisenhardt, 1989). However, in the present study the emergence of unanticipated findings was what led toward studying multilevel strategizing. Therefore, the needed breadth of data to produce analytical generalization can very well hide the new and interesting findings that might emerge from the data if a closer look was taken. In this case study, the depth of the inquiry enabled us to reorient the study to a new direction. Therefore, these kinds of unique phenomena also warrant further research and can open up new avenues for theoretical development.

CONCLUSION

Based on our study, it appears that strategizing occurs on two levels, namely individual actor level and actor group level, and may decelerate innovation diffusion within industrial networks. More specifically, actor groups try to defend their existence when new actors and ways to create value emerge. This can hinder the diffusion of innovations in a network; despite different actor groups having the knowledge of how new value could be created. Furthermore, actors within the actor groups may compete with each other and therefore hinder the diffusion of innovations.

In addition to the theoretical contributions discussed so far, this study contributes to the discussion on using systematic combining as an approach. This point is further explicated in the subsequent “Lessons Learned” section.

Managerially, our research brings three main implications. First, one should understand the group one operates in and the direction that value system is heading toward. This enables assessment of whether participation in a certain group enables value creation when the value system changes. Second, one should reassess who are emerging or potential competitors, and form ties to actors in the main competing group to draw synergistic benefits. This brings competitors into your own group and secures your own role in the value system. Third, firms should constantly reassess how the focal company creates value, what is the role of other network actors in value creation, and how the actors overall contribute to the value creation of the network now and in the future. Collectively this knowledge could be applied to positioning oneself in relation to emerging industry standards, or to building relations that ensure value creation over time.

This research has limitations. First, to better understand network dynamics, longitudinal studies are needed to fully understand how such groups emerge and evolve. Second, our research focuses on incrementally changing value system and thus testing these findings in other types of systems, such as emerging value systems with radical changes (Möller & Svahn, 2003), would bring new insight into how actors and actor groups strategize, and how innovations diffuse.

LESSONS LEARNED

We encourage business-to-business (B2B) researchers to use systematic combining as an approach when the motive is to understand deeper

structures that underlie B2B phenomena. Rather than having an initial framework or testable hypotheses, as suggested by positivistically oriented case study methods (Eisenhardt, 1989; Yin, 2003), an initial idea of the studied phenomena and context is required. In this study innovation diffusion was the initial phenomenon and advertising networks the context.

If the initial idea does not provide empirical or theoretical findings, the researcher has two options. First, the researcher may alter the study by changing the initial idea and consider the phenomena from a purely different perspective. Second, the researcher may look for unanticipated findings that question the original idea, and reorient the study accordingly. Either way, this requires the matching of empirical and theoretical domains. In our study, the unanticipated findings led us to approximate toward a new research problem that was better at uncovering the deeper structures in the network.

The level and unit of analysis should be loosely defined when entering the field. This helps the researcher to find the case from the context. In our study, the initial focus shifted from studying the facilitation and adaptation of innovation into studying the strategizing of intermediary actors and its role in innovation diffusion.

Reporting the process of systematic combining should be made transparent and explicit. Showing the path through which the systematic combining process led to the final research objective and outcome, can help the reader to understand why such an approach was used and how the systematic combining process was executed. In our study, we report this process as a figure depicting the temporal ordering of stages in the methodology section and provided descriptions of each of the stages.

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